

NIRUTHI EDUCATIONAL INSTITUTE · JEE PREPARATION PROGRAMME

DHEERAJ'S JEE TEST

Joint Entrance Examination — Full Mock Test | Paper I

DATE	DURATION	MAX. MARKS	TOTAL QUESTIONS	EXAM TYPE
24 May 2026	3 Hours	300	75	JEE MAIN

Name: Dheeraj

Roll No.:

Batch:

GENERAL INSTRUCTIONS

- The question paper consists of **75 questions** divided equally across **Mathematics, Physics and Chemistry** (25 each).
- Each subject has **20 Multiple Choice Questions (MCQ)** and **5 Numerical Answer Type** questions.
- Use **blue/black ball-point pen** only. Rough work on the space provided at the end.
- Answers once marked cannot be changed. Ensure accuracy before marking.
- For MCQs, choose the single best answer. For Numerical, write the exact integer/decimal value.

+4
Correct answer (MCQ)**-1**
Wrong answer (MCQ)**+4**
Correct (Numerical)**0**
Wrong / Unattempted (Numerical)**① MATHEMATICS**

Sets • Trigonometry • Algebra • Coordinate Geometry • Probability • Calculus

Questions: 1 – 25

Marks: 100 | Part A: MCQ (Q1–20) | Part B: Numerical (Q21–25)

Part A — Multiple Choice Questions (Each carries +4 / -1)

Q1 to Q20

- If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A \cap B$ is:
(A) $\{1, 2\}$ (B) $\{3, 4\}$
(C) $\{5, 6\}$ (D) $\{1, 6\}$
- The value of $\sin^2\theta + \cos^2\theta$ is:
(A) 0 (B) 1
(C) 2 (D) Depends on θ
- The roots of $x^2 - 5x + 6 = 0$ are:
(A) 2, 3 (B) 1, 6
(C) -2, -3 (D) 5, 6
- The number of ways to arrange 5 different books on a shelf is:
(A) 25 (B) 60
(C) 120 (D) 240
- The coefficient of x^2 in $(1 + x)^4$ is:
(A) 4 (B) 6
(C) 12 (D) 24
- The 10th term of the AP: 2, 5, 8, ... is:
(A) 29 (B) 30
(C) 31 (D) 32

7. The sum of the first 4 terms of the GP: 2, 4, 8, ... is:
(A) 28 (B) 30
(C) 32 (D) 34
8. Slope of the line passing through (1, 2) and (3, 6) is:
(A) 1 (B) 2
(C) 3 (D) 4
9. Probability of getting a head when tossing a fair coin is:
(A) 0 (B) $1/4$
(C) $1/2$ (D) 1
10. The negation of the statement "All birds can fly" is:
(A) No birds can fly (B) Some birds cannot fly
(C) All birds cannot fly (D) Some birds can fly
11. If $i = \sqrt{-1}$, then i^8 equals:
(A) 1 (B) -1
(C) i (D) $-i$
12. The middle term in the expansion of $(x + 1)^6$ is:
(A) $15x^3$ (B) $20x^3$
(C) $15x^2$ (D) $20x^2$
13. $\sin 90^\circ$ equals:
(A) 0 (B) 1
(C) -1 (D) $1/2$
14. Mean of 2, 4, 6, 8 is:
(A) 4 (B) 5
(C) 6 (D) 7
15. Equation of line with slope 2 passing through the origin is:
(A) $y = 2x$ (B) $x = 2y$
(C) $y = x + 2$ (D) $2x + y = 0$
16. The value of 5C_2 is:
(A) 5 (B) 10
(C) 15 (D) 20
17. Sum of the first 5 natural numbers is:
(A) 10 (B) 15
(C) 20 (D) 25
18. Probability of getting an even number when a die is rolled is:
(A) $1/6$ (B) $1/3$
(C) $1/2$ (D) $2/3$

19. Domain of $f(x) = 1/(x - 2)$ is:

(A) $x \neq 2$

(B) $x = 2$

(C) $x > 2$

(D) All real numbers

20. Distance of point (3, 4, 12) from the origin is:

(A) 12

(B) 13

(C) 14

(D) 15

Part B — Numerical Answer Type (Each carries +4 / 0)

Q21 to Q25

21. Find the sum of roots of the equation $x^2 - 7x + 10 = 0$.

Write your numerical answer in the box below.

22. Find the 15th term of the AP: 3, 7, 11, ...

Write your numerical answer in the box below.

23. Find the coefficient of x^3 in the expansion of $(1 + x)^5$.

Write your numerical answer in the box below.

24. Find the slope of the line joining the points (2, 3) and (6, 11).

Write your numerical answer in the box below.

25. A coin is tossed twice. What is the total number of possible outcomes?

Write your numerical answer in the box below.



② P H Y S I C S

Mechanics • Gravitation • Waves • Oscillations • Properties of Matter • Thermodynamics

Questions: 26 – 50 (numbered 1–25 within section)

Marks: 100 | Part A: MCQ (Q1–20) | Part B: Numerical (Q21–25)

Part A — Multiple Choice Questions (Each carries +4 / -1)

Q1 to Q20

1. Dheeraj measured the length of a classroom in Dharmavaram as 12.340 m. The number of significant figures is:
(A) 3 (B) 4
(C) 5 (D) 6
2. Harsha rides his bike from Kadiri to Dharmavaram covering 60 km in 2 hours. His average speed is:
(A) 20 km/h (B) 30 km/h
(C) 40 km/h (D) 60 km/h
3. Chintu throws a cricket ball vertically upward with speed 20 m/s. Maximum height reached is approximately: ($g = 10 \text{ m/s}^2$)
(A) 10 m (B) 20 m
(C) 30 m (D) 40 m
4. Maneesh Anna pushes a cart of mass 5 kg with a force of 20 N. The acceleration produced is:
(A) 2 m/s^2 (B) 4 m/s^2
(C) 5 m/s^2 (D) 10 m/s^2
5. A wooden block remains at rest on a rough surface because:
(A) No force acts on it (B) Static friction balances the applied force
(C) Kinetic friction acts on it (D) Gravity disappears
6. Nandhu rides a scooter of total mass 100 kg at 10 m/s. The kinetic energy is:
(A) 1000 J (B) 2000 J
(C) 5000 J (D) 10000 J
7. Harsha lifts a 50 N box by 2 m in 5 s. The power developed is:
(A) 10 W (B) 20 W
(C) 25 W (D) 50 W
8. A spinning fan continues rotating after power is switched off due to:
(A) Newton's first law (B) Conservation of angular momentum
(C) Friction (D) Gravity
9. The approximate value of g near Earth's surface is:
(A) 5 m/s^2 (B) 9.8 m/s^2
(C) 15 m/s^2 (D) 20 m/s^2
10. Escape velocity from Earth is approximately:
(A) 7.9 km/s (B) 11.2 km/s
(C) 15 km/s (D) 20 km/s

11. The restoring force in a spring is proportional to displacement according to:
(A) Newton's law (B) Hooke's law
(C) Ohm's law (D) Kepler's law
12. Time period of a simple pendulum depends on:
(A) Mass of the bob (B) Amplitude
(C) Length of the pendulum (D) Shape of the bob
13. Wave speed is given by:
(A) $v = f / \lambda$ (B) $v = f\lambda$
(C) $v = \lambda / f$ (D) $v = f^2\lambda$
14. A car accelerates uniformly from 0 to 20 m/s in 5 s. Its acceleration is:
(A) 2 m/s^2 (B) 4 m/s^2
(C) 5 m/s^2 (D) 10 m/s^2
15. The dimensional formula of force is:
(A) MLT^{-1} (B) ML^2T^{-2}
(C) MLT^{-2} (D) M^0LT^{-2}
16. Torque is defined as:
(A) Force $\times$ distance (B) Mass $\times$ acceleration
(C) Force $\times$ perpendicular distance (D) Momentum $\times$ velocity
17. Resonance occurs when the frequency of the external force becomes equal to:
(A) Zero (B) Natural frequency of the system
(C) Double the natural frequency (D) Half the natural frequency
18. Pressure is defined as:
(A) Force $\times$ Area (B) Force / Area
(C) Area / Force (D) Mass $\times$ Volume
19. Frequency is the reciprocal of:
(A) Amplitude (B) Velocity
(C) Time period (D) Displacement
20. A freely falling object converts:
(A) Kinetic energy to potential energy (B) Potential energy to kinetic energy
(C) Heat energy to light energy (D) Mass to energy

Part B — Numerical Answer Type (Each carries +4 / 0)

Q21 to Q25

21. Chinna runs 100 m in 20 s. Find his speed in m/s.

Write your numerical answer in the box below.

22. Find the force required to accelerate a 4 kg object at 3 m/s^2 .

Write your numerical answer in the box below.

23. A 10 N force moves an object by 5 m. Calculate the work done (in joules).

Write your numerical answer in the box below.

24. Find the weight of a 6 kg object on Earth. ($g = 10 \text{ m/s}^2$)

Write your numerical answer in the box below.

25. A pendulum completes 20 oscillations in 40 seconds. Find its time period (in seconds).

Write your numerical answer in the box below.



③ CHEMISTRY

Atomic Structure • Bonding • Equilibrium • Thermodynamics • Organic Chemistry • Environmental Chemistry

Questions: 51 – 75 (numbered 1–25 within section)

Marks: 100 | Part A: MCQ (Q1–20) | Part B: Numerical (Q21–25)

Part A — Multiple Choice Questions (Each carries +4 / -1)

Q1 to Q20

1. Amma is boiling water for tea. If 18 g of water is taken, the number of moles of water present is:
(A) 0.5 (B) 1
(C) 2 (D) 18
2. Dheeraj inflated a balloon in summer. The balloon expanded mainly because:
(A) Pressure decreases (B) Temperature increases gas volume
(C) Mass of gas increases (D) Gravity decreases
3. Amma cooks rice in a pressure cooker because the increased pressure:
(A) Decreases the boiling point (B) Increases the boiling point
(C) Stops evaporation completely (D) Permanently converts water into steam
4. When the concentration of reactants is increased, the equilibrium shifts:
(A) Backward (B) Forward
(C) Reaction stops (D) No effect
5. Rusting of iron is an example of:
(A) Reduction (B) Oxidation
(C) Neutralization (D) Combustion
6. Atomic number of an element represents the number of:
(A) Neutrons (B) Electrons only
(C) Protons (D) Mass number
7. Atomic radius generally decreases across a period because:
(A) Atomic mass decreases (B) Nuclear charge increases
(C) Electrons disappear (D) Number of shells decreases
8. In H_2 molecule, bond formation occurs by:
(A) Transfer of electrons (B) Sharing of electrons
(C) Loss of neutrons (D) Gain of protons
9. The shape of the water (H_2O) molecule is:
(A) Linear (B) Trigonal planar
(C) Bent (V-shaped) (D) Tetrahedral
10. Hydrogen gas is:
(A) The heaviest gas (B) The lightest gas
(C) A coloured gas (D) Liquid at room temperature
11. Sodium is stored in kerosene because it:
(A) Is very expensive (B) Reacts vigorously with air and moisture
(C) Is radioactive (D) Evaporates quickly

12. LPG used in Amma's kitchen mainly contains:
- (A) Methane (B) Propane and Butane
(C) Hydrogen (D) Carbon monoxide
13. The alcohol functional group is represented by:
- (A) $-\text{CHO}$ (B) $-\text{OH}$
(C) $-\text{COOH}$ (D) $-\text{NH}_2$
14. Acid rain mainly occurs due to:
- (A) CO_2 (B) SO_2 and NO_x
(C) O_3 (D) Hydrogen gas
15. Hybridization of carbon in methane (CH_4) is:
- (A) sp (B) sp^2
(C) sp^3 (D) dsp^2
16. The ideal gas equation is:
- (A) $PV = nRT$ (B) $E = mc^2$
(C) $F = ma$ (D) $V = IR$
17. Burning LPG in a stove is an example of:
- (A) Endothermic reaction (B) Exothermic reaction
(C) Neutralization reaction (D) Electrolysis
18. At chemical equilibrium:
- (A) The reaction stops completely (B) Forward and backward reactions proceed at equal rates
(C) Only the backward reaction occurs (D) Concentration of products becomes zero
19. The general formula of alkanes is:
- (A) $\text{C}_n \text{H}_{2n}$ (B) $\text{C}_n \text{H}_{2n+2}$
(C) $\text{C}_n \text{H}_{2n-2}$ (D) $\text{C}_n \text{H}_n$
20. The most electronegative element is:
- (A) Oxygen (B) Nitrogen
(C) Fluorine (D) Chlorine

Part B — Numerical Answer Type (Each carries +4 / 0)

Q21 to Q25

21. Calculate the number of moles present in 44 g of CO_2 . (Molar mass of $\text{CO}_2 = 44 \text{ g/mol}$)

Write your numerical answer in the box below.

22. If 100 J of work is done on a system and its internal energy increases by 50 J, find the heat supplied (in joules).

Write your numerical answer in the box below.

23. Pressure of a gas doubles at constant temperature. The final volume becomes what fraction of the original volume?

Write your numerical answer in the box below.

24. Find the valency of oxygen in H_2O .

Write your numerical answer in the box below.

25. What is the number of carbon atoms present in one molecule of propane (C_3H_8)?

Write your numerical answer in the box below.

* E n d o f Q u e s t i o n P a p e r *